COMPUTATION OF THH OF THE ADAMS SUMMAND

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Abstract. These notes are for the talk in the eCHT Kan seminar during 2025 fall. We introduce the Adams splitting of p-local connective complex K-theory and explain how to use Adams spectral sequence to compute the topological Hochschild homology of the Adams summand ℓ. We will also briefly introduce some related topics.

Introduction

The aim of this talk is to explain McClure-Staffeldt's computation of $\mathrm{THH}(\ell)$ in [MS93]. Here THH denotes the topological Hochschild homology. It is the original point of other variants of cyclic homology, which classically are nice approximations of K theory stuffs. The spectrum ℓ is the Adams summand of $\mathrm{ku}_{(p)}$ for a fix prime $p > 2$.

There are many reasons for us to consider the computation for THH. First of all, the topological Hochschild homology is the base for the other players (TC, TC^- and TP) in cyclic worlds, while these stuffs profoundly associate with algebrac K-theory. For example, the well-known DGM theorem says that

Theorem 0.1 ([DGM12]; Theorem 7.2.2.1)**.** *Let $E \to F$ be a map of $\mathbb{E}_\infty$-ring spectra which is surjective on π_0 with a nilpotent kernel. The trace map induces an equivalence*

$$\mathrm{Ker}(K(E) \to K(F)) \xrightarrow{\simeq} \mathrm{Ker}(\mathrm{TC}(E) \to \mathrm{TC}(F))$$

The cyclic world can connect to other arithmatic objects as well. Starting from the comparison between classical Hochschild homology and de Rham cohomology, one may form various comparison maps in arithmatic. For example, we have traditional equivalence.

$$\mathrm{H}(\mathrm{HH}_*(R)) \simeq \Omega^*_{R/\mathbb{Z}}$$

where R is smooth over $\mathbb{Z}$. The later thread of this connection has witnessed a shift of focus from the spectrum to the filtration. Due to Ben Antieau [Ant19], there is an equivalence

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{mot}}\mathrm{HC}^-(R) \simeq \mathrm{fil}^*_{\mathrm{Hdg}}\widehat{\mathrm{dR}}_{R/\mathbb{Z}}$$

where the left hand side is the "double sheering" of the motivic filtration on HC^- and the right hand side is the usual complete Hodge filtration of de Rham complex.

The latest outcome came after [Wag] and [Dev25]. In the former thesis the author constructed the comparison equivalence between the q-Hodge completion of the Hodge filtration of q-de Rham complex and the double sheering filtration of TC^-.

Theorem 0.2 ([Wag]; Theorem 7.9)**.** *There is an equivalence for $p > 2$.*

$$\mathrm{fil}^*_{\mathrm{qHdg}}q\text{-}\widehat{\mathrm{dR}}_{R/A} \xrightarrow{\simeq} \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{ev}}\mathrm{TC}^-_{\blacksquare}(\mathrm{ku}_R/\mathrm{ku}_A)$$

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The consequence is based on the latter thesis, where Devalapurkar proved that there is an $S^1 \times \mathbb{Z}_p^\times$-equivariant equivalence of cyclotomic $\mathbb{E}_\infty$-rings.

$$\mathrm{THH}(\mathbb{Z}_p[\zeta_p]/\mathbb{S}_p[\![q-1]\!]) \simeq \tau_{\geqslant 0}\mathrm{ku}_p^{tC_p}$$

Unfortunately, although the application of THH is fruitful, in general these homology are difficult to compute. Historically, before McClure-Stuffeldt's computation, the only clear result in this direction is Bökstedt's computation of $\mathbb{F}_p$.

Theorem 0.3. *There is an equivalence* $\mathrm{THH}(\mathbb{F}_p) \simeq \mathbb{F}_p[\Omega S^3]$.

Philosophically, the main difficulty in the computation of THH is a general issue that exists in algebraic topology world. That is, it is in general very hard to determine the differentials of spectral sequences. However, if the base spectrum has some periodicity, the associated spectral sequence is more likely to be computed, since we need only to determine the differentials near the zeroth degree, then everything will be determined automatically. This is the main strategy of McClure-Staffeldt's paper. We will explain how to exploit the properties of the Adams spectral sequence of $\mathrm{THH}(\ell)$, and how to use these properties to determine $\pi_*(\mathrm{THH}(\ell); \mathbb{Z}/p)$ eventually.

These notes are divided as follows. In the preliminary section we will first recall the basic facts about Adams splitting of the p-local connective K theory. We will also recall the Adams spectral sequence and the definition of THH. After that we will first determine the second page of the Adams spectral sequence of $\mathrm{THH}(\ell)$, then we will determine the differentials with the help of some $K(1)$-localizations. Finally, we will briefly make some remarks about miscellaneous consequences.

1. Preliminary

In this section, we recall some basic definitions and facts that will be needed in the outline of the computation. All the references are listed so we will omit most of the proof.

1.1. **Adams summand of p-local complex K theory.** Let R be a subring of $\mathbb{Q}$. We use $A(R)$ to denote the set of morphisms $\alpha : K(X;R) \to K(X;R)$, which is natural and R-linear. When $R_1 \subseteq R_2$, we have a monomorphism $A(R_1) \to A(R_2)$. Therefore, we could move our discussion to the settings where $R = \mathbb{Q}$. Using Chern classes, we have identification

$$K(X;\mathbb{Q}) \cong \prod_{n\geqslant 0} \mathrm{H}^{2n}(X;\mathbb{Q})$$

Now we fix a prime $p > 2$, we construct a natural transformation ϕ_k for every $1 \leqslant k \leqslant p-1$ which defines an element in $A(\mathbb{Q})$. More explicitly, we have

$$\phi_k : \prod_{n\geqslant 0} \mathrm{H}^{2n}(X;\mathbb{Q}) \to \prod_{n\geqslant 0} \mathrm{H}^{2n}(X;\mathbb{Q})$$
$$\phi_k(a_0, a_2, \cdots) = (b_0, b_2, \cdots)$$

where

$$b_{2i} = \begin{cases} a_{2i} & \text{for } i = n(p-1)+k \\ 0 & \text{otherwise} \end{cases}$$

Proposition 1.1 ([Ada06]). *The operators ϕ_k lie in $A(\mathbb{Z}_{(p)})$ for all k, where $\mathbb{Z}_{(p)}$ denote the p-localization of integers.*

Proposition 1.2. *The operators ϕ_k enjoy following properties:*

(1) $(\phi_k)^2 = \phi_k$
(2) $\phi_k\phi_j = 0$ *for* $j \neq k$
(3) $\Sigma_k\phi_k = \mathrm{id}$
(4) (Cartan formula) $\phi_k(xy) = \Sigma_{i+j=k}(\phi_i x)(\phi_j y)$

Using these properties, we could split the complex K-theory with coeffitience in $\mathbb{Z}_p$ into $p-1$ factors.

Corollary 1.3. *We have a natural direct sum splitting*

$$K(X;R) = \bigoplus_{1\leqslant i\leqslant p-1} K_i(X)$$

with each $K_i(X) \cong \phi_i K(X;\mathbb{Z}_{(|)})$ a representable functor, and with $K_i(X)K_j(X) \subseteq K_{i+j}(X)$. Therefore, we have a spectrum-level splitting

$$\mathrm{ku}_{(p)} \simeq \bigvee_{0\leqslant i\leqslant p-2} \ell_i$$

Remark 1.4. It is easy to find that all summands in the splitting of Corollary 1.3 are equivalent up to a shift. More explicitly, one can compute that the above splitting could be written as

$$\mathrm{ku}_{(p)} \simeq \bigvee_{0\leqslant i\leqslant p-2} \ell_i \simeq \bigvee_{0\leqslant i\leqslant p-2} \Sigma^{2i}\ell$$

where $\pi_*(\ell) = \ell_* = \mathbb{Z}_{(p)}[u]$ with a single generator in degree $2p-2$. We call the spectrum ℓ the **Adams summand** of $\mathrm{ku}_{(p)}$.

Remark 1.5. One can generalize this splitting from the p-local cases to the p-complete cases. There is a similar splitting result

$$\mathrm{ku}_p \simeq \bigvee_{0\leqslant i\leqslant p-2} \Sigma^{2i}\ell_p$$

where $\pi_*(\ell_p) = \ell_{p*} = \mathbb{Z}_p[u]$ with a single generator in degree $2p-2$. The spectrum ℓ_p is a model for the first Morava K-theory $K(1)$. Moreover, the Adams operator ψ^{p-1} acts permutatively on the splitting, which inherits ku_p a $\mathbb{Z}_p^\times$-action.

1.2. **Adams spectral sequence.** Adams spectral sequence is our main ingredient for calculation. It is also a powerful tool in understanding the homotopy groups. In this subsection we recall the basic construction of Adams spectral sequence.

Definition 1.6. We say a ring spectrum E is of **Adams type** if it satisfies the following conditions

(1) E is a filtered colimit of the form $E \simeq \mathrm{colim}_\alpha E_\alpha$, with each term E_α finite projective over E, i.e. E_*E_α is finitely generated and projective as an E_*-module.
(2) For each α, the E-cohomology of E_α is dual to its E-homology. That means

$$E^*E_\alpha \simeq \mathrm{Hom}_{E_*}(E_*E_\alpha, E_*)$$

One can show that for E of Adams type, E_*E is flat over E_* and $E_*(E\otimes X)\simeq E_*E\otimes_{E_*}E_*X$.

Example 1.7. The examples of Adams type spectra include $\mathbb{Q}$, $\mathbb{F}_p$, ku_p, MU, Lubin-Tate E-theory $E(n)$ and Morava K-theory $K(n)$. $\mathbb{Z}$, however, is not of Adams type, because the integral homology of $\mathbb{Z}$ itself is not flat over $\mathbb{Z}$.

Now we fix an Adams type ring spectrum E and any spectrum Y. We consider the fiber

$$\overline{E}\longrightarrow\mathbb{S}\xrightarrow{e}E$$

We could use this fiber sequence to give Y a filtration. We then have a Postnikov tower

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & \overline{E}^{\otimes 2}\otimes Y & \longrightarrow & \overline{E}\otimes Y & \longrightarrow & Y & \longrightarrow & Y \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & E\otimes\overline{E}^{\otimes 2}\otimes Y & & E\otimes\overline{E}\otimes Y & & E\otimes Y & &
\end{array}$$

Now we consider another spectrum X. After applying $[X,-]$, one can get exact triangles.

$$\begin{array}{ccccccccc}
\cdots & \longrightarrow & [X,\overline{E}^{\otimes 2}\otimes Y] & \longrightarrow & [X,\overline{E}\otimes Y] & \longrightarrow & [X,Y] & \longrightarrow & [X,Y] \\
 & & \downarrow & \swarrow & \downarrow & \swarrow & \downarrow & & \\
 & & [X,E\otimes\overline{E}^{\otimes 2}\otimes Y] & & [X,E\otimes\overline{E}\otimes Y] & & [X,E\otimes Y] & &
\end{array}$$

In this way, we obtain the first page of **E-Adams spectral sequence**

$$E_1^{s,t}=[X,E\otimes\overline{E}^{s}\otimes Y]_{t-s}$$

We look closer at the first page of this spectral sequence.

Proposition 1.8 ([Ada06])**.** *Let E be of Adams type, the natural map*

$$E_*(E)\otimes_{\pi_*(E)}E_*(X)\to\pi_*(E\otimes E\otimes X)$$

induced by

$$\mathrm{id}\otimes\mu\otimes\mathrm{id}:E\otimes E\otimes E\otimes X\to E\otimes E\otimes X$$

is an equivalence. This carrys E_*X a E_*E comodule structure over Steenrod algebra (π_*E,E_*E), which is given by

$$E\otimes X\simeq E\otimes\mathbb{S}\otimes X\xrightarrow{\mathrm{id}\otimes e\otimes\mathrm{id}}E\otimes E\otimes X$$

The functor $E_*(-):\mathrm{Sp}\to\mathrm{Mod}^{\mathbb{Z}}_{\pi_*E}$ in this way lifts to a functor $E_*(-):\mathrm{Sp}\to\mathrm{CoMod}_{E_*E}$. In fact, one can show that when E is of Adams type, $[X,E\otimes Y]\xrightarrow{E_*(-)}\mathrm{Hom}(E_*X,E_*(E\otimes Y))$ is an isomorphism. With this in hand, we have

Theorem 1.9. *Let E be of Adams type, then the first page of E-Adams spectral sequence is isomorphic to*

$$E_1^{s,t}\cong\mathrm{Hom}^t(E_*X,E_{*-s}(E\otimes\overline{E}^{s}\otimes Y))$$

Proof. We have

$$\begin{aligned} E_1^{s,t} &\cong [X, E \otimes \overline{E}^s \otimes Y]_{t-s} \\ &= \mathrm{Hom}^{t-s}(E_*X, E_*(E \otimes \overline{E}^s \otimes Y)) \\ &= \mathrm{Hom}^{t}(E_*X, E_{*-s}(E \otimes \overline{E}^s \otimes Y)) \end{aligned}$$

□

Moreover, one can show that the differentials given by

$$\cdots \to E_*(E \otimes \overline{E}^s \otimes Y) \to E_{*-1}(E \otimes \overline{E}^{s+1} \otimes Y) \to \cdots$$

endow E_*Y with an injective resolution in CoMod_{E_*E}. This could be used to compute the second page of Adams spectral sequence.

Corollary 1.10. *Let E be of Adams type. The second page of the Adams spectral sequence is of the form*

$$E_2^{s,t} = \mathrm{Ext}^{s,t}_{E_*E}(E_*X, E_*Y)$$

where s is the "derived " degree and t is the grading degree.

In this talk, we will restrict to the case where $E = \mathbb{F}_p$. In this case we have the following convergent theorem.

Theorem 1.11 ([Bou79])**.** *Let $E = \mathbb{F}_p$, Y be a connective spectrum and X be a finite spectrum, the Adams spectral sequence* ***converges completely*** *to $[X, Y]^\wedge_p$.*

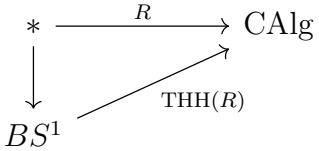
1.3. **Topological Hochschild homology.** We now briefly introduce the topological Hochschild homology. The main reference for this part is [NS18]. We begin with the definition.

Definition 1.12. Let R be a $\mathbb{E}_\infty$ ring spectrum, we define the **topological Hochschild homology** of R to be

$$\mathrm{THH}(R) = R \otimes_{R \otimes R^{op}} R$$

The ring spectrum $\mathrm{THH}(R)$ admits a natural S^1 action. To see this, we need to notice that $S^1 \simeq * \sqcup_{* \sqcup *} *$. In fact, the following theorem shows that the functor $\mathrm{THH}(X)$ is somehow a "free" S^1 equivariant spectrum generated by X.

Theorem 1.13 (McClure-Schwänzel-Vogt)**.** *The functor* $\mathrm{THH} : \mathrm{CAlg} \to \mathrm{CAlg}^{BS^1}$ *is the left Kan extension of the constant map.*



Therefore, one has equivalence

$$\mathrm{Map}_{\mathrm{CAlg}}(R, -) \simeq \mathrm{Map}_{\mathrm{CAlg}^{BS^1}}(\mathrm{THH}(R), -)$$

The computation of THH is not a long story. The first computable example is given by Bökstedt.

Theorem 1.14 ([Bök85])**.** *There is an equivalence of $\mathbb{E}_1$-rings*

$$\mathrm{THH}(\mathbb{F}_p) \simeq \mathbb{F}_p[\Omega S^3]$$

Therefore, we have $\mathrm{THH}(\mathbb{F}_p)_* \simeq \mathbb{F}_p[u]$ with the generator u in degree 2.

Bökstedt also introduced the following spectral sequence for computation.

Theorem 1.15. *For any generalized homology theory K that satisfies the Künneth fomula, one has a spectral sequence*

$$\hat{E}_2 = \mathrm{HH}_*(K_*R) \Rightarrow K_*(\mathrm{THH}(R))$$

Moreover, if we denote the standard map $R \to \mathrm{THH}(R)$ by ι, an element $a \in \hat{E}_2$ detects the element $\iota_*(a) \in K_*(\mathrm{THH}(R))$

Remark 1.16. When the generalized homology theory K no longer satisfies the Künneth fomula, the spectral sequence still exists, where the second page is $K_*(R) \otimes_{K_*(R\otimes R)} K_*(R)$ instead. This is isomorphic to $\mathrm{HH}_*(K_*R)$ if and only if K satisfies the Künneth fomula.

2. Determine the E_2 page

We fix a prime $p \geqslant 3$. In order to calculate $\mathrm{THH}(\ell)_* = \pi_*\mathrm{THH}(\ell)$ with coefficience in $\mathbb{F}_p$, naturally we choose to endow $\mathrm{THH}(\ell)$ an $\mathbb{F}_p$-resolution. In this section, we will compute the second page of the associated Adams spectral sequence.

Let $E_2(\mathrm{THH}(\ell); p)$ be the second page of the $\mathbb{F}_p$-Adams spectral sequence of $\mathrm{THH}(\ell)$. The standard map $\iota : \ell \to \mathrm{THH}(\ell)$ induces a map of spectral sequence

$$\iota_* : E_2(\ell; p) \longrightarrow E_2(\mathrm{THH}(\ell); p)$$

By [**e**], we have already got the following computation.

$$E_2(\ell; p) \cong \mathbb{Z}/p[a_1] \cong E_\infty(\ell; p)$$

with a_1 in degree $(2p-2, 1)$ detecting $v_1 \in \pi_{2p-2}(\ell; p)$ the generator. We are going to compute the following result.

Theorem 2.1. *The second page $E_2(\mathrm{THH}(\ell); p)$ is of the form*

$$\mathbb{Z}/p[a_1] \otimes \mathbb{Z}/p[\mu] \otimes \Lambda[\lambda_1, \lambda_2]$$

where $\mathbb{Z}/p[a_1]$ is identified with the image of ι_*, μ locates in degree $(2p^2, 0)$, λ_1 locates in degree $(2p-1, 0)$, and λ_2 locates in degree $(2p^2-1, 0)$.

The proof strategy is as follows. We have already known that the $\mathbb{Z}/p$-homology of ℓ embeds to A_*, which is the dual Steenrod algebra. Therefore, we will first compute $\hat{E}_2 = \mathrm{HH}_*(\mathrm{H}_*(\ell; \mathbb{Z}/\mathrm{p}))$ which converges to $\mathrm{H}_*(\mathrm{THH}(\ell); \mathbb{Z}/p)$, then we analyze the second page of Adams spectral sequence.

2.1. **Calcultation of $\mathrm{H}_*(\mathrm{THH}(\ell);\mathbb{Z}/p)$.** In order to compute this homology ring. We will use the spectral sequence from Theorem 1.15. We will first prove a general lemma, which helps us to determine a wide range of HH_*.

Lemma 2.2. *If R has the form $\mathbb{Z}/p[x_1,x_2,\cdots]\otimes\Lambda(y_1,y_2,\cdots)$, then $\mathrm{HH}_*(R)$ has the form*

$$R\otimes\Lambda(\sigma(x_1),\sigma(x_2),\cdots)\otimes\Gamma(\sigma(y_1),\sigma(y_2),\cdots)$$

where $\sigma:\mathrm{HH}_0(R)\to\mathrm{HH}_1(R)$ is the suspension and Γ is a divided power series.

Proof. By definition, we have

$$\mathrm{HH}_*(R)=R\otimes^{\mathrm{L}}_{R\otimes R}R\simeq R\otimes^{\mathrm{L}}\mathbb{Z}/p\otimes^{\mathrm{L}}_{R}\mathbb{Z}/p\simeq R\otimes^{\mathrm{L}}\mathrm{Tor}^R(\mathbb{Z}/p,\mathbb{Z}/p)$$

where the computation of $\mathrm{Tor}^R(\mathbb{Z}/p,\mathbb{Z}/p)$ is a standard homological algebra exercise. □

We already know that $\mathrm{H}_*(\ell;\mathbb{Z}/p)$ can be identified with the subalgebra of $\mathrm{H}_*(\mathbb{Z}/p;\mathbb{Z}/p)\cong A_*$, which is

$$\mathbb{Z}/p[\xi_1,\xi_2,\cdots]\otimes\Lambda(\tau_2,\tau_3,\cdots)$$

Applying the above lemma to the situation by hand, we get the second page of the spectral sequence

$$\hat{E}_2(\ell)=\mathrm{HH}_*(\mathrm{H}_*(\ell;\mathbb{Z}/\mathrm{p}))\cong\mathrm{H}_*(\ell;\mathbb{Z}/\mathrm{p})\otimes\Lambda(\sigma(\xi_1),\sigma(\xi_2),\cdots)\otimes\Gamma(\sigma(\tau_2),\sigma(\tau_3),\cdots)$$

The differentials can be computed in $\hat{E}_2(\mathbb{Z}/p)$, which by Bökstedt's computation the only nontrivial differential is d^{p-1}. Following [Bök85], we obtain

$$\hat{E}_p(\ell)\cong\hat{E}_\infty(\ell)\cong\mathrm{H}_*(\ell;\mathbb{Z}/\mathrm{p})\otimes\Lambda(\sigma(\xi_1),\sigma(\xi_2))\otimes\tau_pP(\sigma(\tau_2),\sigma(\tau_3),\cdots)$$

where τ_pP denotes the truncated polynomial of height p.

We can use Dyer-Lashof operation to determine the multiplicative structure. Consequently, one has

$$\mathrm{H}_*(\mathrm{THH}(\ell);\mathbb{Z}/p)\cong\mathrm{H}_*(\ell;\mathbb{Z}/\mathrm{p})\otimes\Lambda(\lambda_1,\lambda_2)\otimes\mathbb{Z}/p[\mu]$$

where λ_i, $i=1,2$ is detected by $\sigma(\xi_i)$; μ is detected by $\sigma(\tau_2)$.

Finally, by Milnor's calculation, one can decide the A_*-comodule structure on $\mathrm{H}_*(\mathrm{THH}(\ell);\mathbb{Z}/p)$. Combining everything together, we obtain

Theorem 2.3. *The homology ring $\mathrm{H}_*(\mathrm{THH}(\ell);\mathbb{Z}/p)$ is of the form*

$$\mathrm{H}_*(\ell;\mathbb{Z}/p)\otimes\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2]$$

where μ is located in degree $(2p^2,0)$, λ_1 is located in degree $(2p-1,0)$, and λ_2 is located in degree $(2p^2-1,0)$, $\mathrm{H}_*(\ell;\mathbb{Z}/p)$ is identified with the image of ι_*.

2.2. **Computation of the second page.** Now we are going to prove Theorem 2.1. This could be obtained immediately from Theorem 2.3. However, in McClure-Staffeldt's paper, they obtained this by analyzing the comodule base change formula.

Proof. The Hopf algebra quotient of the embedding $\mathrm{H}_*(\ell;\mathbb{Z}/p) \to \mathrm{H}_*(\mathbb{Z}/p;\mathbb{Z}/p) \cong A_*$ can be identified with $\Sigma = \Lambda(\tau_1)$, which follows from the above discussion of homology rings. Therefore, we have

$$\begin{aligned} E_2(\mathrm{THH}(\ell)) &= \mathrm{Ext}_{A_*}(\mathbb{Z}/p;\mathrm{H}_*(\mathrm{THH}(\ell);\mathbb{Z}/p)) \\ &\cong \mathrm{Ext}_{A_*}(\mathbb{Z}/p;\mathrm{H}_*(\ell;\mathbb{Z}/p)\otimes\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2]) \\ &\cong \mathrm{Ext}_{\Sigma}(\mathbb{Z}/p;\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2]) \\ &\cong \mathrm{Ext}_{\Sigma}(\mathbb{Z}/p;\mathbb{Z}/p)\otimes\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2] \end{aligned}$$

where the third equation is from the base change formula for comodules, and the last equation is because the coaction on $\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2]$ is trivial □

3. Determine the differentials

The strategy for us to decide the differentials is to embed E_r in its localization $a^{-1}E_r$. In order to do this, we need some "formal language" in the spectrum level. More explicity, we will use $K(1)$-localization of spectrum.

Theorem 3.1. *Let $K(1)$ denote the first Morava K-theory. The map ι is $K(1)$-equivalence. i.e. The inclusion $\iota : \ell \to \mathrm{THH}(\ell)$ induces an isomorphism*

$$\iota_* : K(1)_*\ell \xrightarrow{\simeq} K(1)_*\mathrm{THH}(\ell)$$

Proof. This can be checked using the Bökstedt spectral sequence from Theorem 1.15. □

Corollary 3.2. *The map ι induces an isomorphism*

$$v_1^{-1}\pi_*(\ell;\mathbb{Z}/p) \xrightarrow{\cong} v_1^{-1}\pi_*(\mathrm{THH}(\ell);\mathbb{Z}/p)$$

Proof. Theorem 3.1 implies immediately that $L_{K(1)}\ell \simeq L_{K(1)}\mathrm{THH}(\ell)$. Therefore, we obtain the isomorphism by the following square.

$$\begin{array}{ccc} v_1^{-1}\pi_*(\ell;\mathbb{Z}/p) & \xrightarrow{\cong} & v_1^{-1}\pi_*(\mathrm{THH}(\ell);\mathbb{Z}/p) \\ \cong\downarrow & & \cong\downarrow \\ \pi_*(L_{K(1)}\ell;\mathbb{Z}/p) & \xrightarrow[\cong]{} & \pi_*(L_{K(1)}\mathrm{THH}(\ell);\mathbb{Z}/p) \end{array}$$

where the vertical arrow is given by [Bou79] □

We have computed that the second page of the Adams spectral sequence is

$$E_2(\mathrm{THH}(\ell)) = \mathbb{Z}/p[a_1]\otimes\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2]$$

We use localization to invert the single element a_1. Then we get a new spectral sequence, which we denote by $a_1^{-1}E_r(\mathrm{THH}(\ell))$. The Adams spectral sequence of $\mathrm{THH}(\ell)$ embeds in $a_1^{-1}E_r(\mathrm{THH}(\ell))$. Especially, we have

$$a_1^{-1}E_2(\mathrm{THH}(\ell)) = \mathbb{Z}/p[a_1,a_1^{-1}]\otimes\mathbb{Z}/p[\mu]\otimes\Lambda[\lambda_1,\lambda_2]$$

We must be careful because we said nothing about the convergence about this new "localized" spectral sequence. However, one can always factor everything smoothly with the help of the comparison map.

Theorem 3.3. *There is an equivalence induced by $\iota : \ell \to \mathrm{THH}(\ell)$.*

$$\hat{\iota}_* : \mathbb{Z}/p[a_1, a_1^{-1}] \cong a_1^{-1}\pi_*(\ell; \mathbb{Z}/p) \xrightarrow{\cong} a_1^{-1}E_\infty(\mathrm{THH}(\ell))$$

Proof. We will give a detailed check.

First, a_1^n cannot be in the kernel, since by Corollary 3.2, v_1^n is a non-zero element in $\pi_*(\mathrm{THH}(\ell); \mathbb{Z}/p)$ and the only element that can detect v_1^n is a_1^n, which cannot be zero.

On the other hand, suppose that x is in the cokernel of the above comparison map. After being multipicated by a power of a_1 we may assume that $x \in E_\infty(\mathrm{THH}(\ell))$ and $\xi \in \pi_*(\mathrm{THH}(\ell); \mathbb{Z}/p)$ detected by x. However, again by inspecting the v_1-localized homotopy ring of $\mathrm{THH}(\ell)$ one shows that $\xi - v_1^n$ must be killed by a power of v_1. This implies that x is killed by some power of a_1, which contradicts the assumption. □

Now we can check the differentials in $a_1^{-1}E_r(\mathrm{THH}(\ell))$.

Theorem 3.4. *Let $a_1^{-1}E_r(\mathrm{THH}(\ell))$ be as above.*

(1) The only non-trivial differentials of $a_1^{-1}E_r(\mathrm{THH}(\ell))$ is located in $r(n)$-th page, where $r(n), n \geqslant 1$ is defined as

$$r(n) = \begin{cases} p + p^3 + \cdots + p^{n-2} + p^n & \text{if } n \text{ is odd} \\ p^2 + p^4 + \cdots + p^{n-2} + p^n & \text{if } n \text{ is even} \end{cases}$$

(2) $a^{-1}E_{r(n)}(\mathrm{THH}(\ell))$ has the form

$$a_1^{-1}E_r(\mathrm{THH}(\ell)) \cong \mathbb{Z}/p[a_1, a_1^{-1}] \otimes \mathbb{Z}/p[\mu^{p^{n-1}}] \otimes \Lambda[\lambda_n, \lambda_{n+1}]$$

where λ_n is inductively defined as $\lambda_n = \lambda_{n-2}\mu^{p^{n-3}(n-2)}$.

(3) The differentials are fully determined by Leibnitz rule and the following relations

$$d_{r(n)}(\lambda_n) = 0 \quad d_{r(n)}(\lambda_{n+1}) = 0 \quad d_{r(n)}(\mu^{p^{n-1}}) = ma_1^{r(n)}\lambda_n$$

where m is an element in $\mathbb{Z}/p^\times$

Proof. We first consider elements λ_1 and λ_2. We claim that these two elements must be permanent cycles. This is because the $(2p-2)$-th colume contains no elements, so λ_1 will hit nothing. The only nontrivial element in the $(2p^2-2)$-th colume is a_1^{p+1}. But λ_2 cannot hit a_1^{p+1}, otherwise a_1 will die, which contradicts Theorem 3.3.

Now we analyze the element μ, the only nontrivial element in the $(2p^2-1)$-th colume is $a_1^p\lambda_1$, which lies in bidegree $(2p^2-1, p)$. This guarantees that μ at least survives to the p-th page. However, μ cannot survive to the end because it cannot detect any elements, similarly by Theorem 3.3.

Therefore, we must have $d_p(\mu) = ma_1^p\lambda_1$ for some non-zero m. Since a_1 is invertible all the time, λ_1 vanishs in the next page. By Leibnitz rule, one can compute that $d_p(\mu^2) = 2ma_1^p\lambda_1\mu$ and so on. We can continue till we reach $d_p(\mu^p)$, where we find that d_p becomes zero again because of the base field we choose guarantees that $p = 0$. Consequently, we obtain

$$a_1^{-1}E_{p+1}(\mathrm{THH}(\ell)) = \mathbb{Z}/p[a_1, a_1^{-1}] \otimes \mathbb{Z}/p[\mu^p] \otimes \Lambda[\lambda_2, \lambda_1\mu^{p-1}]$$

We can go on similar discussion for μ_p. By induction, we arrive at the consequence. □

By checking the survival of a_1 in $E_r(\mathrm{THH}(\ell))$, we obtain the following consequence, which allows us to compare and determine the differentials in $E_r(\mathrm{THH}(\ell))$ completely by those of $a^{-1}E_{r(n)}(\mathrm{THH}(\ell))$.

Proposition 3.5. *For each r, the differentials of $E_r(\mathrm{THH}(\ell))$ are fully determined by that of $a^{-1}E_{r(n)}(\mathrm{THH}(\ell))$. Every element in $E_r(\mathrm{THH}(\ell))$ has the same differentials as its image in $a^{-1}E_{r(n)}(\mathrm{THH}(\ell))$.*

We can finish the computation now.

Corollary 3.6. *The homotopy ring $\pi_*(\mathrm{THH}(\ell);\mathbb{Z}/p)$ is generated as a $\mathbb{Z}/p[a_1] \cong \pi_*(\ell;\mathbb{Z}/p)$-module by elements $x_{m,n}$; $\quad n \geqslant 1; m \neq p-1 \mod p$ and relationship $v_1^{r(n)}x_{n,m} = v_1^{r(n)}x'_{n,m} = 0$. The elements $x_{m,n}$ are detected by $\lambda_n\mu^{mp^{n-1}}$ and $x'_{m,n}$ are detected by $\lambda_n\lambda_{n+1}\mu^{mp^{n-1}}$.*

4. Miscellaneous consequences

In this section, we make some remarks about the miscellaneous results in McClure and Staffeldt's article.

Remark 4.1. The similar method can be used to compute the homotopy group of THH(BP). The procedure to compute this is totally the same. We first apply Bökstedt's spectral sequence to compute the homology

$$\mathrm{H}_*(\mathrm{THH}(\mathrm{BP});\mathbb{Z}/p) \cong \mathrm{H}_*(\mathrm{BP};\mathbb{Z}/p) \otimes \Lambda(\xi_1,\xi_2,\cdots)$$

Then using the same argument for Hopf algebra, we obtain the second page of the Adams spectral sequence and one can compute immediately that

$$\pi_*\mathrm{THH}(\mathrm{BP}) \cong \pi_*\mathrm{BP} \otimes \Lambda(\lambda_1,\lambda_2,\cdots)$$

Remark 4.2. One can also show that we have the equivalence of spectrum, where ℓ_p is the p-adic Adams summand.

$$\mathrm{THH}(\ell_p) \simeq \ell_p \vee L_{\mathbb{Q}}\Sigma\ell_p$$

To see this, one first form the global pullback square along profinite completion

$$\begin{array}{ccc} \mathrm{THH}(\ell_p) & \longrightarrow & \prod_q \mathrm{THH}(\ell_p)^\wedge_q \\ \downarrow & & \downarrow \\ L_{\mathbb{Q}}\mathrm{THH}(\ell_p) & \longrightarrow & L_{\mathbb{Q}}\big(\prod_q \mathrm{THH}(\ell_p)\big) \end{array}$$

One can show that we have equivalences

$$\ell_p \simeq \mathrm{THH}(\ell_p)^\wedge_p$$

$$L_{\mathbb{Q}}\ell_p \simeq L_{\mathbb{Q}}\big(\prod_q \mathrm{THH}(\ell_p)^\wedge_q\big)$$

Therefore, one can check the claim by considering after rationalization.

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